Moving the Factory to the Field: GIS-Optimized Autothermal Mobile Pyrolysis Units for Converting Surplus Crop Residue into Renewable Fuel and Power in Bangladesh

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Category: 1, Generating 10,450 MW from renewable energy by 2030 (waste-to-energy pathway, Strategy Paper §5.4)

Group: A, student team (Team Oratory)

Interactive 3D model of the unit: https://nakib.tech/oratory-model/

1. Problem

Bangladesh burns a fuel it already owns. Every year it produces about 57.4 Mt of net available crop residue, 49.7 Mt of it rice straw [2], enough in principle to support more than 9,000 MW of electricity [3].

Almost none of it reaches energy markets. Straw is bulky, wet and scattered, so hauling it to a central plant costs more than the straw is worth.

Surplus straw is burned in the field or left to rot, while kilns and textile boilers nearby burn imported coal and furnace oil bought in dollars.

The Strategy Paper gives waste only 200 MW of its 10,450 MW target, because central plants rarely pay without tipping fees, carbon credits and by-product sales [1]. Transport is what breaks them.

2. Proposed Solution

Our answer is to move the factory to the residue. Each Oratory unit is a truck-mounted pyrolysis reactor that processes 10 dry tonnes a day at the field edge, at 400–550 °C without oxygen.

The unit is autothermal: once started, it burns its own pyrogas for heat and needs no external fuel. Its layout is shown in an interactive 3D model at https://nakib.tech/oratory-model/ [4].

Table 1 shows what one dry tonne becomes. A truckload of bio-oil carries about ten times the energy of a truckload of loose straw, which removes the transport cost that sinks central plants.

Table 1. Products from one dry tonne of residue.

Product	Yield	Use
Bio-oil	about 550 kg, 16–24 MJ/kg	Replaces furnace oil in industrial boilers and furnaces
Char (“Green Coal”)	about 220 kg	Kiln and boiler fuel, or biochar; 90 kg returned to the farmer
Pyrogas	remainder	Heats the reactor and dries incoming straw, making the unit autothermal

Deployment is GIS-optimized. Satellite imagery and upazila crop statistics locate residue-dense grid cells; units go to their centroids, and network analysis plots the shortest freight routes to buyers.

The fleet follows the harvest: Aman straw in Rangpur and Dinajpur (November–January), Rabi stalks (February–March), Boro straw (April–June) and jute sticks in Faridpur and Jessore (July–August).

3. Contribution to the Category 1 Target

The Strategy targets 20% renewable electricity, about 27,000 GWh, by 2030 [1]. Running 300 days a year, one unit turns 3,000 dry tonnes into about 1,650 t of bio-oil, roughly 28 TJ of energy.

That equals 1.1 MW of continuous thermal output, or 0.3 MW of electricity at 30% generator efficiency. It uses no farmland.

Fifty units by 2030 would process 150,000 t of residue, only 0.3% of national rice straw, and supply about 16 MW of electricity equivalent: roughly 8% of the 200 MW waste-to-energy allocation.

4. Business Model and Unit Economics

The model pays its way without subsidy or carbon revenue. Farmers and village co-operatives are paid for residue, baling and short hauls; industries pay for fuel (Table 2).

Table 2. Estimated costs and revenue per dry tonne of residue (Tk).

Cost item	Tk	Revenue item	Tk
Feedstock, baling, haul	3,000	Bio-oil, 550 kg at Tk 24/kg	13,200
Crew of eight	970	Biochar sold, 130 kg at Tk 12/kg	1,600
Freight and relocation	930	Biochar returned, 90 kg	in kind
Maintenance	730	Carbon credits	not counted
Utilities and overheads	1,030		
Total cost	**6,700**	**Total revenue**	**14,800**

Each tonne earns a margin of Tk 8,100, or about Tk 2.43 crore of operating cash flow per unit a year. At an estimated capital cost of Tk 11 crore, simple payback is about 4.5 years.

Buyers save too. Bio-oil is priced at about half the cost of furnace oil per unit of energy, and its price is set in taka, not dollars.

Market entry. We lead with Green Coal because buyers burn it in equipment they already own: one free trial load to a brick kiln, cement plant or textile mill, then a harvest-season contract.

Bio-oil follows for dyeing and washing mills short of gas.

Financing. We will apply to the Strategy's proposed Renewable Energy Development Fund, and seek Bangladesh Bank refinancing, credit guarantees and SME green finance (§8) [1].

Carbon credits, excluded above, will be registered with the Power Division under Article 6 of the Paris Agreement (§9) [1].

5. Technical Feasibility and Risk Management

The core process is proven. Pyrolysis of crop residue is established; what we add is autothermal mobile operation and GIS-optimized routing.

We plan to fabricate reactor shells, condensers and frames with local heavy engineering workshops, and import only controls and burners. Table 3 lists the main risks.

Table 3. Principal risks and mitigation measures.

Risk	Mitigation
Variable feed moisture	Bales field-dried below 15% moisture; on-board dryer heated by pyrogas; every batch checked
Variable bio-oil quality	Tolerant industrial boilers as first buyers; batch testing; a product specification with BSTI
Competition with fodder	Buy only surplus straw at a fair price; return biochar
Fragmented crop data	Upazila statistics verified by field agents replace real-time routing
High capital cost	Phased fleet growth, financed as described in Section 4

6. Impact and Scalability

Each unit is expected, every year, to:

- keep 3,000 t of residue out of open burning and decay;
- pay about Tk 90 lakh in villages for residue, baling and haulage;

- create eight skilled jobs, plus seasonal work in collection;
- return 270 t of biochar to farmers' fields;
- avoid an estimated 790 t CO_2e (team estimate).

At 50 units, this becomes 150,000 t of residue, about 400 skilled jobs and about 40,000 t CO_2e avoided each year.

In line with the Strategy (§11–12) [1], women will make up at least 30% of trained technicians, and each co-operative will run a complaints channel for farmers.

7. Implementation Plan

- *November 2026 to March 2027:* GIS mapping of one pilot district; detailed design, fabrication and permits.
- *April to June 2027:* pilot unit operates through the Boro harvest; fuel trials with two industrial buyers.
- *2028:* five units in service; carbon credit registration.
- *2029 to 2030:* fleet expands to 50 units, in step with the Strategy's deployment ramp from 2028 [1].

The cost and yield figures above are team estimates, not yet tested in the field. The pilot verifies them before a second unit is built.

8. Policy Support Requested

Three low-cost measures would let the model scale:

1. coverage of mobile units under the Strategy's pyrolysis standards (§5.4);
2. a single national permit for mobile units, in place of approval at every site;
3. access for mobile waste-to-energy units to the Strategy's financing instruments (§8).

References

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